AI Controller Based on RL for Non-Linear Systems

Proposed PI-PPO-DR Framework for inverted Pendulum

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Introduction: Evolution of Control Strategies

Nonlinear control problems like the inverted pendulum require robust strategies. Over time, control approaches have evolved to handle greater complexity:

- **PID Controllers:** Highly effective for local stabilization when the pendulum is already near the upright position ($\theta \approx 0$).
  Limitation: They fail for large initial angles due to reliance on linearization.
- **Deep Q-Networks (DQN):** Enabled global swing-up from any angle.
  Disadvantage: DQN relies on discrete action spaces, resulting in "bang-bang" control. This causes high-frequency chattering, which can severely damage physical actuators.
- **Proximal Policy Optimization (PPO):** **Our chosen approach.**
  Advantage: It natively handles continuous action spaces, allowing for smooth, stable, and wear-free global stabilization.

Understanding PPO

PPO is a state-of-the-art policy gradient method that balances sample efficiency with ease of tuning.

- **Actor-Critic Architecture:** Uses an Actor network to predict continuous actions and a Critic network to evaluate state value.
- **Continuous Output:** The action is drawn from a probability distribution:

$$u_t \sim \mathcal{N}(\mu_\theta(s_t), \sigma_\theta(s_t))$$

- **Clipped Objective:** PPO restricts how much the policy can change in a single update, ensuring monotonic learning and preventing the network from forgetting how to balance.

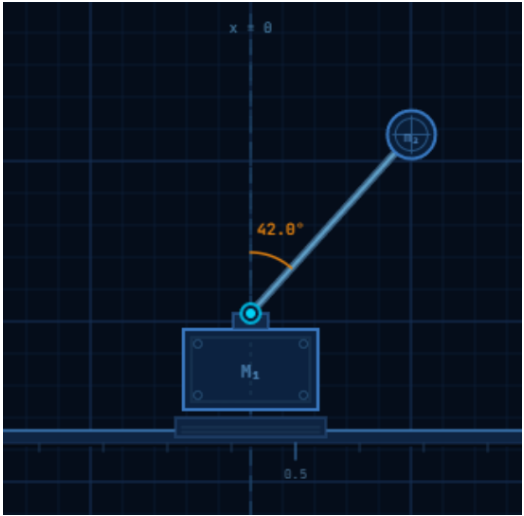
The Proposed Framework: PI-PPO-DR

To address the shortcomings of pure RL, we introduce the **PI-PPO-DR** Framework:

1. **PPO (Proximal Policy Optimization):** Provides the base algorithm for stable, continuous control.
2. **PI (Physics-Informed):** Embeds the system's energy dynamics directly into the reward function to guide the agent efficiently.
3. **DR (Domain Randomization):** Randomizes physical parameters during simulation to ensure the trained policy is robust enough for real-world hardware.

System Overview

- Cart-pendulum system
- Nonlinear dynamic model
- Controlled using RL agent
- Agent outputs continuous force



State Representation

System state:

$$s_t = [x_t, \dot{x}_t, \theta_t, \dot{\theta}_t]$$

Where

- x_t : cart position
- $\dot{x}_t$: cart velocity
- θ_t : pendulum angle
- $\dot{\theta}_t$: angular velocity

Augmented state:

$$s_t^{aug} = [x_t, \dot{x}_t, \theta_t, \dot{\theta}_t, u_{t-1}]$$

Including previous action improves smoothness of control.

Continuous Control via PPO

The control input is a bounded force:

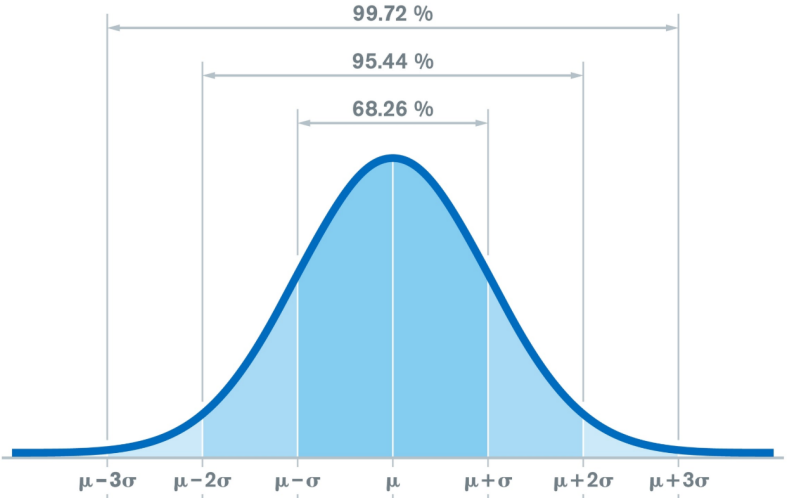
$$u_t \in [-F_{max}, F_{max}]$$

Policy network:

$$u_t \sim \mathcal{N}(\mu_\theta(s_t), \sigma_\theta(s_t))$$

- Mean μ_θ produced by neural network
- Standard deviation controls exploration
- Output clipped to physical limits

Normal Distribution



$$u_t \sim \mathcal{N}(\mu_\theta(s_t), \sigma_\theta(s_t))$$

Domain Randomization & Training Initialization

1. Domain Randomization (Physical Robustness)

To improve sim-to-real transfer, physical system parameters are sampled from a uniform distribution ($\mathcal{U}$) at the start of each episode:

Parameter	Symbol	Nominal Value	Randomization Range
Cart Mass	M_c	1.0 kg	$\mathcal{U}(0.75M_c^{nom}, 1.25M_c^{nom})$
Pendulum Mass	M_p	0.1 kg	$\mathcal{U}(0.75M_p^{nom}, 1.25M_p^{nom})$
Pendulum Length	L	0.5 m	$\mathcal{U}(0.8L^{nom}, 1.2L^{nom})$
Max Force	F_{max}	10 N	$\mathcal{U}(0.9F_{max}^{nom}, 1.1F_{max}^{nom})$
Friction/Damping	b	0.1 Ns/m	$\mathcal{U}(0.7b^{nom}, 1.3b^{nom})$
Gravity	g	9.81 m/s^2	Constant

2. Training Initialization (Global Stabilization)

The initial state is randomized every episode. The starting **pendulum angle, angular velocity, cart position, and cart velocity** are all chosen dynamically from broad uniform distributions. This forces the agent to learn a universal recovery and swing-up strategy from any arbitrary starting orientation.

Physics-Informed Reward: Energy Term

The total reward is a combination of energy, precision, and smoothness components: $R_t = R_{energy,t} + R_{precision,t} + R_{smooth,t}$.

Energy Term (Swing-Up Behavior): Guides energy injection to enable swing-up from downward states.

$$\text{Total Energy: } E_t = \frac{1}{2} M_p L^2 \dot{\theta}_t^2 + M_p g L (1 + \cos\theta_t)$$

$$\text{Desired Upright: } E_{upright} = 2 M_p g L$$

$$\text{Penalty: } R_{energy,t} = -w_E (E_t - E_{upright})^2$$

Physics-Informed Reward: Precision & Smoothness

Precision Stabilization Term: Reduces oscillation, ensures a stable equilibrium, and centers the cart.

$$R_{precision,t} = -(w_\theta \theta_t^2 + w_{\dot{\theta}} \dot{\theta}_t^2) - w_x x_t^2 - w_{\dot{x}} \dot{x}_t^2$$

Smoothness Penalty: Reduces actuator stress and prevents chattering by penalizing large control inputs and sudden changes in force.

$$R_{smooth,t} = -w_u u_t^2 - w_{\Delta u}(u_t - u_{t-1})^2$$

Adaptive Blending & Unified Reward Function

1. Adaptive Blending (α_t)

Smoothly transitions between swing-up (energy control) and balancing (precision control):

$$\alpha_t = \frac{|\theta_t|}{|\theta_t| + \theta_c}$$

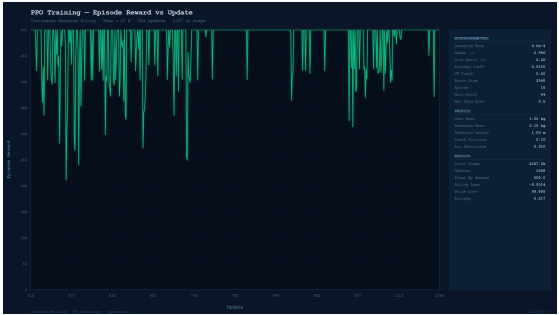
Where the transition parameter is initialized as $\theta_c = 0.3$ *rad.*

2. Final Unified Reward Function (R_t)

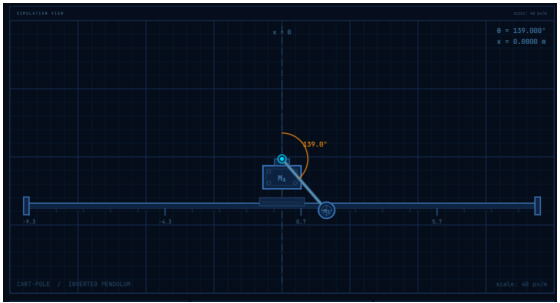
Combines the energy, precision, and smoothness terms using the blending factor:

$$\begin{aligned} R_t = &- \alpha_t w_E (\Delta E_t)^2 \\ &- (1 - \alpha_t)(w_\theta \theta_t^2 + w_{\dot\theta} \dot\theta_t^2) \\ &- w_x x_t^2 - w_{\dot x} \dot x_t^2 \\ &- w_u u_t^2 - w_{\Delta u} (u_t - u_{t-1})^2 \end{aligned}$$

Simulation Results



Training curve for the PPO agent



System with initial angle of 139 degrees

Overall Advantages

This proposed PI-PPO-DR framework offers several key benefits:

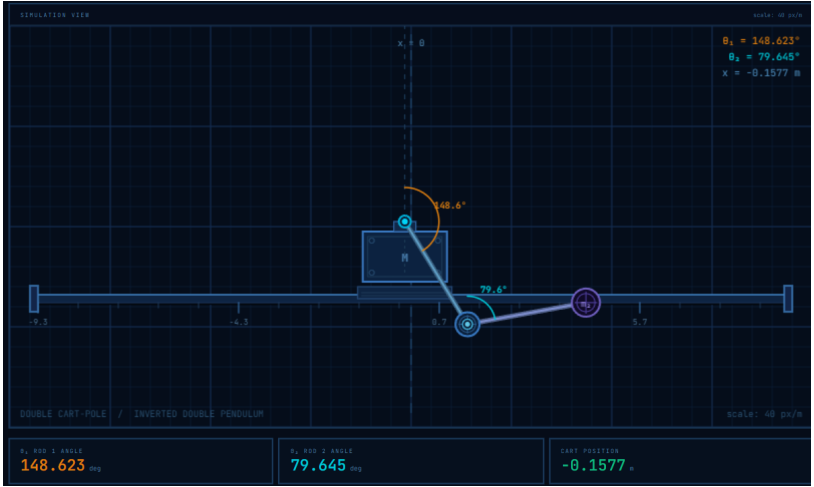
- Enables smooth, continuous control.
- Supports global nonlinear stabilization without needing multiple discrete controllers.
- Integrates fundamental physics knowledge for faster, more guided convergence.
- Reduces actuator stress through action augmentation and smoothness penalties.
- Enhances sim-to-real robustness via comprehensive domain randomization.
- Provides highly structured, easily tunable reward shaping.

Future Work: The Acrobot / Double Pendulum

While the single inverted pendulum demonstrates the framework's viability, our next objective is to scale this approach to more highly nonlinear, underactuated systems.

- **Acrobot System Implementation:** Applying PI-PPO-DR to a double pendulum system where only the middle joint is actuated.
- **Complex Energy Shaping:** Modifying the physics-informed reward to account for the coupled chaotic dynamics of two links.
- **Framework Transferability:** Demonstrating the versatility of the core PI-PPO-DR architecture by adapting it from a single cart-pole environment to an entirely different multi-link topology.

Next Phase



Early-stage development of the Acrobot (Double Inverted Pendulum) simulation environment to test framework transferability.

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Thank You

Questions?